$\therefore$ The condition that for each column containing a leading entry, all positions below it must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 3 = Row 2 Column 3 = 0
   Row 1 Column 2 = 0

$\therefore$ The condition that for each column containing a leading entry, all positions above it must contain 0 is True.

So, the final matrix is in rref $\blacksquare$.

· By defn, each of the leading entries of the first 3 rows are pivots.

Row 4 has no pivots.

$\therefore T: \mathbb{R}^4 \to \mathbb{R}^4$ is not onto.

It's easy to see that the corresponding rref of $\begin{bmatrix} A & 0 \end{bmatrix}$ is

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

There is no row in the matrix of the form $\begin{bmatrix} 0 & 0 & 0 & \ldots & 0 & b \end{bmatrix}$ where $b \neq 0$. By Theorem 2, the corresponding system of equations is consistent.

· By defn, x_4 is a free variable since Column 4 has no pivots.

· By Theorem 2, the matrix equation $Ax = 0$ has infinitely many solutions.

$\boxed{\therefore T: \mathbb{R}^4 \to \mathbb{R}^4 \text{ is not one-to-one}}$ $\blacksquare$

27) The transformation in Exercise 19.

Solution:

$$\begin{bmatrix} 1 & -5 & 4 \\ 0 & 0 & -6 \end{bmatrix} \xrightarrow[\times(-1/6)]{\text{Row1} = \text{Row1}} \begin{bmatrix} 1 & -5 & 4 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[-4 \times \text{Row2}]{\text{Row1} = \text{Row1}} \begin{bmatrix} 1 & -5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Verification of whether the final matrix is in rref

1. Row 1 and 2 are both non-zero rows. $\therefore$ The condition that all the non-zero rows must be above the zero rows is vacuously True.

2. Row 1 Leading entry = 1 in Column 1
   Row 2 Leading entry = 1 in Column 3